BundleFuse: From Prior Work to the Research Question

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This document develops the motivation and positioning before the experimental case. Its purpose is to explain what is inherited, what question is being asked, and how the implemented construction addresses that question. It is a focused positioning argument, not an exhaustive literature review or a claim of established priority. The accompanying technical note defines the exact scores and their assumptions.

1 Begin with the kind of relationship the task requires

A partial bundle specifies a context for choosing another item. A camera bag can complete a camera kit even though its appearance and function differ from the camera. A second camera may be more similar but less useful as a completion. Thus a useful ranking must represent the relationship required by the context, rather than assume that resemblance always implies suitability.

This distinction has an established basis. McAuley, Pandey and Leskovec model substitutable and complementary product links using product-associated information, including review text and category structure [17]. Their work motivates relation-sensitive recommendation. It does not establish that every pair from different categories is complementary, or that a category rule solves compatibility.

For bundle completion, the relationship can also vary by domain. Repetition of a product role can be undesirable in an outfit, while a shared artist can be useful in a playlist. This motivates retaining several precisely defined relationship scores. It does not justify assuming that all of them are useful everywhere.

This distinction is already central to fashion compatibility research. Vasileva et al. learn type-aware embeddings that jointly represent similarity and compatibility [14]. Their method learns a representation from images; BundleFuse instead computes category and descriptive scores. The difference is in the construction, not the recognition that item type matters.

2 What explicit recommendation models already establish

Item-relation methods provide a natural starting point: score a candidate by its relationships to the observed items. Let $\mathcal{I}$ be a catalogue, S its observed subset, $c \notin S$ a candidate, and M_{cs} an estimated relation between c and s. The additive score is

$$r(c \mid S) = \sum_{s \in S} M_{cs}.$$

This notation describes a class of estimators; it is not presented as a new formula.

Sarwar et al. establish the item-neighborhood approach: estimate item-item relations from user ratings and use them to form predictions [12]. BundleFuse applies a related principle to bundle

co-memberships, which record presence rather than explicit ratings. The analogy does not make the two inputs or their meanings identical.

EASE demonstrates that a shallow linear recommendation model with a closed-form solution can achieve strong ranking performance [18]. Side-information extensions of SLIM already combine purchase histories and item attributes in a regularized linear model [19]. More recently, the FEASE preprint extends EASE with user and item side features for cold-start handling [20].

These works constrain our claim. Explicit scoring, side-information fusion and the absence of a deep downstream encoder are established possibilities. Furthermore, a model originally formulated for user histories can often be applied to an observed item set. Merely renaming a history as a partial bundle does not create a new method.

The remaining design issue for this work is more specific: *which relations should an explicit estimator preserve when completing bundles, and how should their scores be combined?* Historical overlap, category relations and descriptive resemblance are related measurements, but they are not interchangeable definitions of compatibility.

3 What modern bundle construction already addresses

CLHE directly addresses sparse bundle histories by combining multimodal features, item-level user feedback and bundle composition. It uses hierarchical self-attention and item- and bundle-level contrastive objectives [21]. Its motivation already includes the observation that features can be available for items with little bundle history. BundleFuse must acknowledge that motivation as prior work.

RaMen develops characteristic and collaborative strategies together with hypergraph structure to learn bundle representations [22]. It explicitly models multiple kinds of relationships. A claim that earlier bundle models consider only similarity, ignore direct item associations, or neglect multi-item structure would misrepresent this comparator.

These papers motivate a question about the *construction of the predictor*. They do not imply that another learned encoder is the only way to use the available information, nor that their representation learning is unnecessary. That issue is comparative and must be investigated.

4 State the question without inventing an unaddressed problem

The research question is:

> Given historical bundles and available item attributes, how far can explicit, relation-specific compatibility estimates support bundle completion without training a new neural bundle encoder?

This asks about the usefulness and limits of a particular alternative construction. It does not assert that no earlier model combines these inputs. Its value depends on defining a nontrivial construction, explaining its choices, and eventually showing what those choices achieve relative to appropriate alternatives.

The distinction from neural bundle encoders is that the relevant relationships enter as directly computed candidate scores. The distinction from a generic linear recommender must be established through the specific relation constructions and combination rule, not merely through the claim that the method is simple.

The combination of history and descriptions also has a long history. Melville et al. use content predictions to enrich sparse user ratings before collaborative filtering [13]. BundleFuse forms

several candidate scores from bundle history and item attributes and then combines them; it does not first fill a user rating matrix. Both inherit the idea that content can help where collaborative observations are sparse, so this idea is not itself our contribution.

An even closer task-level precedent is Volkovs et al.'s playlist-continuation system [15]. Its retrieval stage blends item- and playlist-neighbor scores with other collaborative models, adjusts scores for popularity, and is followed by a learned reranker. BundleFuse's particular relations, scaling and fallback differ, but combining multiple retrieval signals and correcting popularity in a partial-set task are already established. A fair comparison must account for task protocol, inputs, and the learned reranking stage.

5 Develop the construction from that question

5.1 Preserve directly observed relations

Let B be the binary training-bundle by item matrix and let x_S be the indicator vector of S. Then $(B^\top B x_S)_c$ counts recorded co-memberships with the query items. BundleFuse uses weighted overlap and alternative summaries of history to reduce particular effects of frequency, bundle size and sparse direct observations. This step inherits the explicit item-relation perspective. Its particular weights and summary operators are modeling choices, not consequences of the cited papers.

5.2 Share evidence while retaining the type of relationship

Exact co-occurrence has no direct evidence for an unobserved pair. BundleFuse also constructs category relations and scores from descriptive features. Category relations aggregate observations across items, while feature relationships extend coverage beyond recorded pairs. The category component can distinguish support between categories from penalties for repeating a rarely repeated category.

These choices connect the relation-sensitive view to an explicit score construction. They do not identify economic substitutes or complements from category labels alone. The distinction is operational: each formula specifies which observed relationship it measures and what response it assigns to the candidate.

Category or cluster centering of descriptors and graph-adjusted features provide further candidate estimators. Their purpose is to separate a group's mean descriptor from within-group variation and to combine descriptions with historical relationships. These operations are not proofs of disentanglement. Their individual utility remains an empirical question. The complete method should disclose each operation, even when several share one conceptual role.

5.3 Combine scores with different scales and coverage

Let $r_h(c \mid S)$ denote raw channel h, let $z_h(c \mid S)$ be its catalogue-standardized value for the query, and let $w_h \geq 0$ be its validation-selected weight. The additive core is

$$Y_0(c \mid S) = \sum_h w_h z_h(c \mid S).$$

This exposes the construction to inspection: the meaning of a channel, its scale transformation and its contribution to a candidate score are explicit. It does not make channels independent or their weights causally interpretable. The use of validation to choose weights is fitting, even without neural gradient training.

A closer interpretation of the history component comes from set-neighbor recommendation. For binary training bundles b, define $m_b = |b \cap S|$, $a_b = \sum_{s \in b \cap S} (f_s + 5)^{-\alpha}$ and $v_b = (|b| + 1)^{-\gamma}$. Before scale alignment, the pair score plus its incidence correction can be written, for nonnegative β, as

$$\sum_{b: c \in b} v_b a_b [1 + \beta \max(0, m_b - 1)].$$

It is therefore a sum over candidate-containing historical bundles with a nonlinear query-overlap weight. Playlist- and session-neighbor methods already score candidates through similar historical sets [8, 9]. Our particular weighting does not impose a nearest-neighbor cutoff, and its formula is not claimed to be identical to theirs. Weighted-incidence dataset profiles need their additional bundle weights retained explicitly.

Further prior art limits a general novelty claim: normalized hypergraph incidence operators predate this method [10]; playlist hybrids have fused co-occurrence with audio and text [11]. Earlier basket-completion models based on determinantal point processes also model set quality and diversity [3, 4]. A detailed map of these and other overlaps, including unresolved exact-formula questions, is provided in `method_prior_art_review.md`.

The signal menu therefore belongs inside the explanation of these relation estimates. It is not a list of independent novelty claims. A convincing presentation must explain why each family is considered, state its overlap with other families, and identify which components survive selection in the reported model.

5.4 Introduce optional corrections after the core

Let Y be the combined score including any selected multi-item corrections, and μ_0 a scaled fallback assembled from a restricted set of channels. Let f_c be the number of candidate appearances in training and $\kappa > 0$ the fallback strength. The backoff score is

$$\theta(c \mid S) = \frac{f_c}{f_c + \kappa} Y(c \mid S) + \frac{\kappa}{f_c + \kappa} \mu_0(c \mid S).$$

It provides an explicit rule for changing the balance with candidate history. Frequency is a proxy for available historical evidence, not a calibrated uncertainty. The two estimates overlap; the fallback is not a history-independent expert.

Related empirical-Bayes product-search work already combines behavioral information with non-behavioral priors [23]. Consequently, neither interpolation nor shrinkage toward descriptive evidence is new by itself. BundleFuse's specific fallback, its overlap with the combined score, and the candidate-frequency rule must be defined and defended in the bundle setting. The technical note gives a pointwise regularization interpretation without assuming a validated noise model. Hu et al. already use an item-statistics-dependent weight to balance behavioral and content representations [1]. CoHeat combines two user–bundle scores with a bundle-popularity-dependent weight and proves the corresponding monotonic influence properties [2]. The task, scores and weighting function differ from BundleFuse, but the general adaptive-mixture principle and monotonic sensitivity argument are established prior work.

The incidence correction also admits a precise relation to earlier itemset methods. In the binary, unit-bundle-weight case, put $d_s = (f_s + 5)^{-\alpha}$ and $v_b = (|b| + 1)^{-\gamma}$. Expanding its overlap factors gives

$$\Delta(c \mid S) = \beta \sum_{\substack{s,t \in S \\ s \neq t}} d_s \sum_b v_b B_{bc} B_{bs} B_{bt}.$$

Thus the correction aggregates weighted historical co-occurrences of each seed pair with the candidate. Higher-order neighborhood models and HOSLIM already score candidates using

itemset-to-item associations [5]. HOSLIM learns sparse coefficients; this construction uses weighted counts and specific shared weights. The formula does not imply the two predictors are equal. The broader idea of amplifying strongly matching neighbors also predates this construction [6, 7].

The incidence surplus and Rao-based term address an additional question: can support involving several observed items change the assessment beyond separate matches? They are explicit optional corrections. Their presence does not establish that higher-order interactions are the principal explanation of the method's accuracy.

Higher-order item interactions in recommendation also have prior work: DeepICF models interactions among previously consumed items with neural networks [16]. This is a different mechanism from BundleFuse's explicit corrections, not evidence that considering several items jointly is new.

6 State the contribution at the level the construction supports

The proposed methodological contribution is the *particular explicit scoring construction for bundle completion*: historical relation estimates, category and descriptive estimates, query-wise scale alignment, and specified optional corrections. The paper must distinguish those design choices from the established ingredients. A coherent combination can be useful, but naming its ingredients differently does not establish novelty.

The central hypothesis is that carefully constructed relation scores provide a competitive and inspectable alternative for this task. This is an affirmative, testable hypothesis. It does not require asserting that neural models are excessive, that all low-rank representations lose necessary information, or that the scorer has a universal computational advantage.

A reviewer may reasonably ask whether BundleFuse is a feature-engineered ensemble. The descriptive answer is that it combines engineered scores. The scientific answer must specify the relationships those scores encode, why those relationships matter for completion, and the distinctive behavior of the resulting estimator. A new technical principle cannot be inferred solely from benchmark wins. The contribution may ultimately be a well-supported construction and analysis rather than a general new theory of recommendation.

The literature reviewed here does not establish that the complete combination is unprecedented. A priority claim requires a closer component-by-component comparison with the nearest published constructions and their underlying data. An explicit score allows inspection of its arithmetic; it does not by itself establish causal explanations or that the scores match human reasons.

7 An introduction built from this path

Completing a bundle requires selecting items that are appropriate companions to those already present. Prior work on product relationships distinguishes substitutability from complementarity, establishing that relevance to an item need not mean resemblance [17]. For completion, historical co-membership supplies direct relational evidence, while item attributes allow that evidence to extend beyond exact observed pairs.

Both explicit and neural models have established ways to exploit such information. Shallow and side-information recommendation models demonstrate the viability of explicit scoring [18, 19, 20]; modern bundle models learn representations from several sources and relationships [21, 22]. These developments motivate examining the extent to which bundle compatibility can be estimated directly from specified relations, with the contributions of those relations remaining visible.

We investigate this question through BundleFuse. The method constructs historical, category and descriptive compatibility scores, places them on comparable scales, and selects their combination

using validation examples. It also provides explicit candidate-frequency and multi-item corrections. This formulation makes the design question concrete: which relational estimates are useful for completing a bundle, and when do the corrections add information beyond their combination?

How this relates to the existing documents. This is the motivation and positioning that should precede `publication_argument_technical.tex`. The historical manuscript's introduction and related-work text contain stronger claims about cost, zero parameters, prior methods and broad superiority that require reconciliation. They should not be used as the current statement of the contribution. Experimental evidence belongs after this argument and must remain within its verified scope.

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